

Harder Trigonometry Exam Questions 2

Total - 54 Marks (1 hour, 24 minutes)

1. (Sydney Girls 2022 Q2b)

(i) Prove that $(\sin \alpha - \cos \alpha)^2 = 1 - \sin 2\alpha$. 2

(ii) Hence, or otherwise, find the exact value of $\sin 75^\circ - \cos 75^\circ$. Leave your answer with a rational denominator. 2

2. (Caringbah 2025 Q8b)

(a) Find the exact value of $(\sin 22.5^\circ + \cos 22.5^\circ)^2$ 2

(b) If $\sin x = \frac{1}{3}$, $\cos y = -\frac{1}{\sqrt{3}}$ such that $\frac{\pi}{2} < x < \pi$ and $\pi < y < \frac{3\pi}{2}$. 4

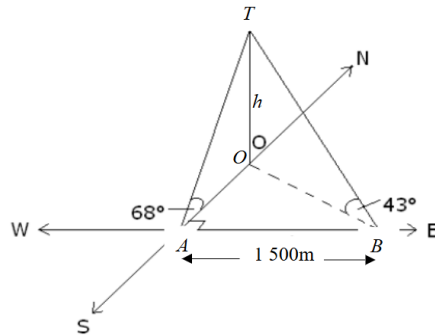
Find the exact value of $\tan(x + y)$.

(c) Prove that $\frac{\sin 3\theta}{\sin \theta} + \frac{\cos 3\theta}{\cos \theta} = 4 \cos 2\theta$ 3

3. (Blacktown Boys 2022 Q24)

Prove that $\tan \theta(1 + \tan \theta) + (1 + \cot \theta) = \frac{\tan \theta + 1}{\sin \theta \cos \theta}$ 3

4. (Hornsby Girls 2021 Y11 Half-Yearly Q11b) A lighthouse of height h metres, is observed from two points, A and B , in the same horizontal plane as the base of the lighthouse. From point A , due south of the lighthouse, the angle of elevation to the top of the lighthouse is 68° . From point B , due east of point A , the angle of elevation to the top of the lighthouse is 43° . Points A and B are 1 500 metres apart.



NOT TO SCALE

(i) Show that $OA = h \tan 22^\circ$ 2

(ii) Show that $h = \frac{1\,500}{\sqrt{\tan^2 47^\circ - \tan^2 22^\circ}}$ 3

(iii) Find the height of the tower correct to the nearest metre. 1

5. (Sydney Boys 2025 Q11c)

Solve $\sin \theta + \cos \theta = -1$ for $0 \leq \theta \leq 2\pi$. 2

6. (Penrith 2022 Q13c)

The equation $I(t) = 4 \sin 0.02t + 8 \cos 0.02t$ measures the current, I amperes, in a particular circuit after t seconds.

(i) Express the function in the form $I(t) = A \sin(at + b)$, for $0 \leq b \leq \frac{\pi}{2}$. 3

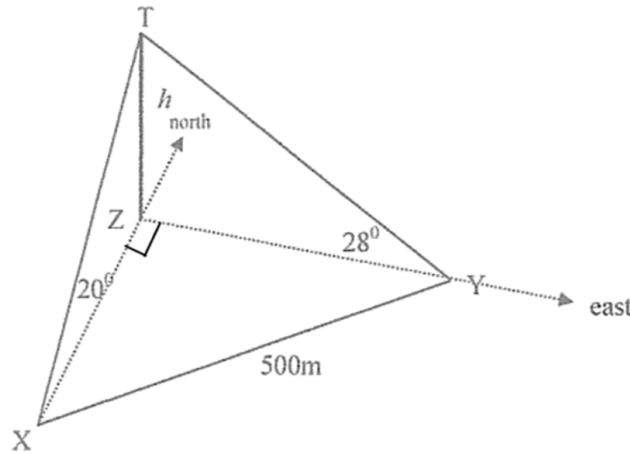
(ii) When does the current first reach its maximum value? 3

7. (Sydney Boys 2020 Q13b)

Solve $\frac{\sin x + \cos x}{\sqrt{2}} = \frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 360^\circ$.

3

8. (Girraween 2023 Preliminary Q12b) A conservationist observes the angle of elevation of the top of a tree, which is h metres tall, from two positions. From a point X , due south of the tree, it is 20° and from point Y , due east of the tree, it is 28° . The distance XY is 500 m.



(i) Write expressions for XZ and YZ in terms of h .

2

(ii) Calculate the value of h .

2

9. (Normanhurst Boys 2019 Q15 a,b& c)

(a) Find the exact value of $(\sin 22.5^\circ + \cos 22.5^\circ)^2$

2

(b) If $\sin x = \frac{1}{3}$, $\cos y = -\frac{1}{\sqrt{3}}$ such that $\frac{\pi}{2} < x < \pi$ and $\pi < y < \frac{3\pi}{2}$.

4

Find the exact value of $\tan(x + y)$.

(c) Prove that $\frac{\sin 3\theta}{\sin \theta} + \frac{\cos 3\theta}{\cos \theta} = 4 \cos 2\theta$

3

10. (James Ruse 2020 Q8a)

(i) Prove the trigonometric identity: $\sin^4 \theta + \cos^4 \theta \equiv \frac{3 + \cos 4\theta}{4}$.

2

(ii) Hence, or otherwise, solve $\sin^4 x + \cos^4 x = \frac{1}{2}$ in the domain $0 \leq x \leq \pi$.

2

11. (James Ruse 2021 Q44)

Given that $m = a(\cos x)^3 + 3a \cos x(\sin x)^2$ and $n = a(\sin x)^3 + 3a \sin x(\cos x)^2$

Prove that $(m + n)^{\frac{2}{3}} + (m - n)^{\frac{2}{3}} = 2a^{\frac{2}{3}}$

4